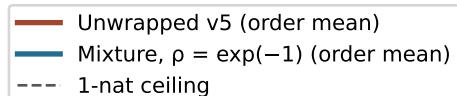


Bounding rare prediction loss at a fixed prefix

Exposed realization-0 evaluation, five orders

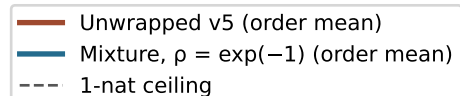
$P(\Delta\text{NLL} > x)$, all 245,237 positions per memory

zsre



Positive loss increase vs base (nats)

counterfact



Positive loss increase vs base (nats)

Cap-off has zero positive loss increase. Curves include zero / beneficial positions in the denominator.
Per-token bound against the same base, not a total-loss or generated-text guarantee. No independent-token uncertainty.